BSIT 375: Knowledge Discovery and Data Science
Lecture Notes: The Comprehensive Data Preprocessing Pipeline

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Preface

These notes synthesize the theoretical foundations of data preprocessing as outlined by Larose & Larose, supplemented with the statistical frameworks required for robust knowledge discovery. We categorize the pipeline into five major functional pillars.

Data Preproessing

Data preprocessing is the process of transforming raw data into a refined format suitable for analysis. Because real-world data is often "dirty", containing errors, missing values, or inconsistent formats, this stage is critical for ensuring model accuracy. The pipeline is generally divided into four major categories:

1 Pillar 1: Data Cleaning

The primary objective of cleaning is to minimize the error term ϵ in the observation $x = x^* + \epsilon$.

1.1 Handling Missing Data

Missingness is not always a random occurrence. Understanding the underlying mechanism is vital for choosing the correct recovery strategy.

1. MCAR (Missing Completely at Random)

- $P(M|Y_{obs}, Y_{mis}) = P(M)$
- **Concept**: The missingness is a purely accidental "act of God." It has no relationship with any value in the dataset, observed or unobserved.
- **Example**: A data entry clerk accidentally skips a row, or a laboratory sample is broken during transport.
- **Treatment**: Listwise or pairwise deletion is generally acceptable as it does not introduce bias (though it reduces power).

2. MAR (Missing at Random)

- $P(M|Y_{obs}, Y_{mis}) = P(M|Y_{obs})$
- **Concept**: The "missingness" is predictable if we look at other variables we *did* collect. It is systematic but observable.
- **Example**: Men are statistically less likely to report their weight than women. We can use the "Gender" column to predict the missing "Weight" values.
- **Treatment**: Advanced imputation such as k-Nearest Neighbors (k-NN), Multiple Imputation, or Regression.

3. MNAR (Missing Not at Random)

- $P(M|Y_{obs}, Y_{mis})$ depends on Y_{mis}
- **Concept**: The data is "hiding." The probability of a value being missing is directly related to the value itself.
- **Example**: High-income earners refusing to disclose their salary in a census, or patients with severe symptoms dropping out of a clinical trial.

- **Treatment**: Highly complex; requires modeling the missingness mechanism specifically or performing sensitivity analysis. Simple imputation often leads to biased results.

2 Noise Smoothing and Outlier Detection

Outlier detection is the process of identifying data points that deviate significantly from the rest of the dataset. These points can represent experimental errors (noise) or rare but valid phenomena. Two primary statistical heuristics are commonly employed:

2.0.1 Z-Score Method (Parametric)

The Z-Score, or standard score, identifies outliers by quantifying how many standard deviations a data point lies from the mean. This method assumes the underlying data follows a **Normal (Gaussian) Distribution**.

- **Mathematical Basis**: The Z-score for a sample x_i is calculated as:

$$z_i = \frac{x_i - \mu}{\sigma}$$

  where μ is the population mean and σ is the standard deviation.

- **Heuristic**: A data point is typically flagged as an outlier if $|z_i| > 3$.
- **Intuition**: According to the *Empirical Rule* ($68 - 95 - 99.7$ rule), 99.7% of data in a normal distribution falls within three standard deviations. Values beyond this threshold have a probability of occurrence of less than 0.3%.
- **Limitation**: It is highly sensitive to extreme values, as the mean and standard deviation themselves are influenced by the outliers they are meant to detect.

2.0.2 IQR Method (Non-Parametric)

The Interquartile Range (IQR) method is a "robust" alternative that does not assume a specific distribution. It focuses on the spread of the middle 50% of the data, making it less sensitive to extreme values.

- **Mathematical Basis**: The IQR is the distance between the first quartile (Q_1, 25th percentile) and the third quartile (Q_3, 75th percentile):

$$IQR = Q_3 - Q_1$$

- **Heuristic**: Data points are flagged if they fall outside the "inner fences":

$$x_i \notin [Q_1 - 1.5 \times IQR,\ Q_3 + 1.5 \times IQR]$$

- **Intuition**: This method was popularized by John Tukey. The multiplier 1.5 is a standard convention; points beyond this range are considered "mild" outliers, while a multiplier of 3.0 is often used to identify "extreme" outliers.
- **Advantage**: Because it relies on medians and quartiles rather than the mean, it is far more stable (robust) in the presence of skewed data or heavy-tailed distributions.

2.1 Redundancy and Correlation Analysis

In data integration, redundancy occurs when an attribute can be "derived" from another. This often leads to multi-collinearity, which can degrade the performance of machine learning models. We employ different statistical measures depending on the data type.

2.1.1 Pearson Correlation Coefficient (Numerical Data)

The Pearson Correlation Coefficient (r) measures the strength and direction of the linear relationship between two continuous variables.

Description The coefficient ranges from -1 to $+1$:

- $r > 0$: Positive correlation (as A increases, B increases).
- $r < 0$: Negative correlation (as A increases, B decreases).
- $r = 0$: No linear correlation.

Numerical Example Suppose we want to check the correlation between *Hours Studied* (A) and *Exam Score* (B) for 3 students:

Student	Hours (a_i)	Score (b_i)
1	2	50
2	4	70
3	6	90

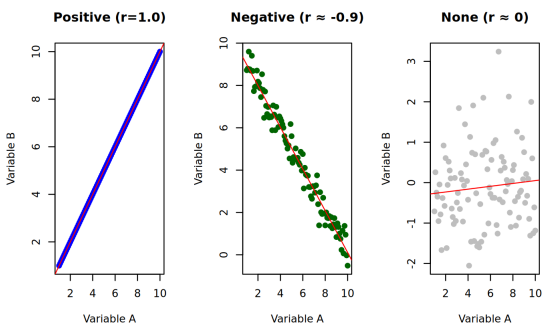
1. **Means**: $\bar{a} = 4$, $\bar{b} = 70$.
2. **Standard Deviations**: $s_A \approx 2$, $s_B = 20$.
3. **Covariance Calculation**:

$$\sum(a_i - \bar{a})(b_i - \bar{b}) = (2-4)(50-70) + (4-4)(70-70) + (6-4)(90-70) = 40 + 0 + 40 = 80$$

4. **Correlation**:

$$r_{A,B} = \frac{80}{(3-1) \times 2 \times 20} = \frac{80}{80} = 1.0$$

An $r = 1.0$ indicates a perfect positive linear redundancy; one of these attributes could likely be removed.



2.1.2 Chi-Square (χ^2) Test (Categorical Data)

For nominal/categorical attributes, we test the hypothesis of independence. If two attributes are highly dependent, they are redundant.

Description The χ^2 test compares the **Observed** frequencies (O) in a contingency table against the **Expected** frequencies (E) that would occur if the variables were completely independent. The expected value for a cell is:

$$E = \frac{(\text{row total} \times \text{column total})}{\text{grand total}}$$

Numerical Example Consider two attributes: `Gender` (Male/Female) and `Preference` (Online/Offline Shopping).

	Online	Offline	Total
Male	20 (Obs)	30 (Obs)	50
Female	40 (Obs)	10 (Obs)	50
Total	60	40	100

1. **Calculate Expected (E) for Male-Online**: $E = \frac{50 \times 60}{100} = 30$.
2. **Calculate Expected (E) for Female-Online**: $E = \frac{50 \times 40}{100} = 20$.
3. **Compute χ^2 term-by-term**:

$$\chi^2 = \frac{(20-30)^2}{30} + \frac{(30-20)^2}{20} + \frac{(40-30)^2}{30} + \frac{(10-20)^2}{20}$$

$$\chi^2 = 3.33 + 5.0 + 3.33 + 5.0 = 16.66$$

If 16.66 exceeds the critical value from the χ^2 distribution table (at a specific significance level like 0.05), we conclude the attributes are strongly related (redundant).

3 Pillar 3: Data Transformation

Data transformation involves converting data from its raw format into a state that enhances the performance and accuracy of analytical models. This phase addresses issues of scale and the mathematical representation of non-numeric data.

3.1 Normalization Techniques

Normalizing features ensures that the objective function of a model (e.g., the cost function in Gradient Descent) treats all dimensions with equal weight.

3.1.1 Min-Max Scaling (Rescaling)

Min-Max scaling "squishes" the data into a fixed range, typically $[0, 1]$.

- **Formula**: $x' = \frac{x - \min(x)}{\max(x) - \min(x)}$
- **Use Case**: Best used when the distribution of data is not Gaussian and you need a bounded range (e.g., for Image Processing or Neural Networks).

- **Drawback**: It is highly sensitive to outliers. If a single extreme value exists, the "normal" data points will be compressed into a very tiny range (e.g., $[0, 0.01]$), losing their descriptive power.

3.1.2 Standardization (Z-score Normalization)

Standardization centers the data around a mean (μ) of 0 with a standard deviation (σ) of 1.

- **Formula**: $z = \frac{x-\mu}{\sigma}$
- **Use Case**: Essential for algorithms that assume a Gaussian distribution, such as Linear Regression, Logistic Regression, and Linear Discriminant Analysis (LDA).
- **Advantage**: Unlike Min-Max, Standardization does not have a bounding range and is significantly more robust to outliers. Even with extreme values, the mean remains relatively stable.

3.2 Categorical Encoding

Since most mathematical models cannot interpret strings like "Red" or "High," we must map categorical labels to numeric values.

3.2.1 One-Hot Encoding (Nominal Data)

For categories with no inherent order (e.g., Color: Red, Blue, Green), we create k binary columns for k levels.

- **Example**: A "City" column with [Kathmandu, Pokhara, Lalitpur] becomes three columns where a "1" represents the presence of that city and "0" represents its absence.
- **The Dummy Variable Trap**: To avoid multi-collinearity, we often drop one column ($k-1$ encoding). If we know it is not Kathmandu and not Pokhara, it must be Lalitpur.
- **Constraint**: This can lead to the "Curse of Dimensionality" if a feature has hundreds of unique categories.

3.2.2 Label Encoding (Ordinal Data)

For categories with a clear rank or order (e.g., Level: Low, Medium, High), we assign a unique integer to each.

- **Example**: Low $\rightarrow$ 1, Medium $\rightarrow$ 2, High $\rightarrow$ 3.
- **Caution**: Never use Label Encoding for nominal data. If you encode "Apple" as 1 and "Banana" as 2, the model might mathematically assume that "Banana" is twice as much as "Apple," which is nonsensical.

Table 1: Comparison of Encoding Techniques

Feature	**One-Hot Encoding**	**Label Encoding**
Data Type	Nominal (No Order)	Ordinal (Has Order)
Dimensionality	Increases (k columns)	Stays same (1 column)
Algorithm Risk	Multi-collinearity	Improper Ordinal Bias

4 Pillar 4: Data Reduction

Data reduction techniques are applied to obtain a reduced representation of the data set that is much smaller in volume, yet closely maintains the integrity of the original data. This improves efficiency and reduces the "Curse of Dimensionality."

4.1 Dimensionality Reduction

Dimensionality reduction reduces the number of random variables under consideration by obtaining a set of principal variables.

4.1.1 Principal Component Analysis (PCA)

PCA is a linear transformation technique that identifies the "directions" (principal components) along which the variance in the data is maximal.

- **Mechanism**: It transforms the original n coordinates into a new coordinate system such that the first coordinate (PC1) captures the highest variance, the second (PC2) the second highest, and so on.
- **Orthogonality**: All principal components are orthogonal (perpendicular) to each other, meaning they are linearly uncorrelated. This effectively eliminates multi-collinearity.
- **Benefit**: By keeping only the top k components, we can represent high-dimensional data in 2D or 3D for visualization while retaining most of the information.

4.2 Numerosity Reduction

Numerosity reduction replaces the original data volume with alternative, smaller forms of data representation. It is categorized into parametric and non-parametric methods.

4.2.1 Parametric Reduction (Regression and Log-Linear Models)

In parametric methods, the data is estimated using a model. Instead of storing the actual data points, only the model parameters are stored.

- **Linear Regression**: Data is modeled to fit a straight line $y = wx + b$. We store only the weight (w) and intercept (b).
- **Log-Linear Models**: Used to approximate the discrete probability distribution of multi-dimensional categorical data.

4.2.2 Non-Parametric Reduction

These methods do not assume a specific functional form.

- **Histograms**: Data is partitioned into bins, and the average or sum of each bin is stored. This is highly effective for approximating data distributions.
- **Clustering**: The dataset is partitioned into clusters. In a reduced representation, each cluster is represented by its **centroid** (mean) or a representative object (medoid).
- **Sampling**: A small random sample (s) is used to represent the entire population (N). Common techniques include *Simple Random Sampling* and *Stratified Sampling* (to ensure sub-groups are represented).

Table 2: Comparison of Reduction Strategies

Strategy	**Approach**	**Key Benefit**
Dimensionality	Feature Extraction (PCA)	Eliminates redundant attributes.
Numerosity	Data Approximation	Reduces storage and CPU load.
Sampling	Representative Subset	Faster prototyping and EDA.

5 Pillar 5: Data Discretization

Data discretization is a form of data reduction that replaces raw continuous values with interval labels or conceptual ranks. This simplifies the data description and is a prerequisite for many rule-based machine learning algorithms.

5.1 Binning Techniques

Binning is a top-down discretization strategy that partitions values into "buckets" or "bins."

5.1.1 Equal-Width (Distance) Binning

This method divides the range of the variable into N intervals of equal size.

- **Formula**: The width of each interval is $w = \frac{\max(x)-\min(x)}{N}$.
- **Example**: Consider sorted prices: $\{5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215\}$. Range $= 215 - 5 = 210$. For $N = 3$, width $= 70$.
  - Bin 1: $[5, 75]$ (Contains: 5, 10, 11, 13, 15, 35, 50, 55, 72)
  - Bin 2: $(75, 145]$ (Contains: 92)
  - Bin 3: $(145, 215]$ (Contains: 204, 215)
- **Observation**: This method is easy to implement but can be heavily skewed if there are outliers (as seen in Bin 2 having only one element).

5.1.2 Equal-Frequency (Depth) Binning

This method ensures that each bin contains approximately the same number of data points.

- **Mechanism**: If we have n samples and want N bins, each bin contains n/N samples.
- **Example**: Using the same prices ($n = 12$), for $N = 3$, each bin must have $12/3 = 4$ elements.
  - Bin 1: $\{5, 10, 11, 13\}$
  - Bin 2: $\{15, 35, 50, 55\}$
  - Bin 3: $\{72, 92, 204, 215\}$
- **Observation**: This is more "robust" to skewed data and ensures that the model sees an equal representation of each interval.

5.2 Concept Hierarchies

Concept hierarchies allow for discretization by transforming low-level "primitive" data into higher-level concepts.

5.2.1 Hierarchical Aggregation

Instead of mathematical intervals, we use domain knowledge to group data.

- **Categorical**: *Kathmandu* $\rightarrow$ *Bagmati Province* $\rightarrow$ *Nepal* $\rightarrow$ *South Asia*.
- **Numerical**: *Age 12* $\rightarrow$ *Child*; *Age 25* $\rightarrow$ *Adult*; *Age 70* $\rightarrow$ *Senior*.

This process, known as **Data Generalization**, reduces the number of distinct values while keeping the semantics intact.

Table 3: Comparison of Discretization Methods

Method	**Optimization Goal**	**Risk**
Equal-Width	Maintains value distribution	Sensitive to outliers
Equal-Frequency	Maintains sample distribution	May group very different values
Entropy-Based	Maximizes Class Purity	Requires target labels (Supervised)